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ORIGINAL ARTICLE

Toll-like receptors 2 and 4 in renal ischemia/reperfusion injury

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Abstract Toll-like receptors (TLRs) are an evolutionarily conserved family of cell membrane receptors that are part of the innate immunity system playing an important role as a first response to tissue injury. TLR2 and TLR4 are constitutively expressed on renal epithelium, and their expression is enhanced following renal ischemia/reperfusion (I/R) injury. Genetic deletion of either TLR2 or TLR4 protects from renal I/R injury. However, it is not known whether deletion of both combined protects the kidney more than a deletion of either one alone. Therefore, we performed renal I/R injury in mice lacking TLR2, TLR4, and TLR2/4, respectively. Our results demonstrate that there are no significant differences regarding protection from renal I/R injury in TLR2/4$^{(-/-)}$ compared with either TLR2$^{(-/-)}$ or TLR4$^{(-/-)}$ gene-targeted mice as determined by histological evaluation and renal functional parameters. Furthermore, there was no difference in the number of apoptotic tubular cells and in nuclear translocation of nuclear factor kappa-B (NF-κB) between the TLR-gene-targeted groups. In parallel, in vitro experiments did not demonstrate an additional effect of the double genetic deletion compared with the single gene deletion with respect to tumor necrosis factor (TNF)-alpha and interleukin (IL)-8 production in hypoxic isolated proximal tubular epithelial cells of the respective animals. In conclusion, a double genetic deletion of TLR2 and TLR4 confers a similar protection following renal I/R injury compared with single deletions of TLR2 and TLR4.

Keywords Kidney · Ischemia · Transplantation · Innate immunity · Toll-like receptors · Knock-out mice · Cell culture

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Introduction

Ischemia/reperfusion (I/R) injury is characterized by inflammation as well as cell death and has a significant clinical impact in the setting of renal transplantation [1]. Direct toxic effects of hypoxia mediated by reactive oxygen species (ROS) induce apoptosis/necrosis during early stages of I/R injury. Furthermore, inflammatory processes are triggered after this initial event, leading to a variable degree of further organ damage. Understanding the complete pathogenesis of immune activation in the setting of I/R injury is only emerging. Evidence shows that the innate immune system with its Toll-like receptors (TLRs) influences the development of I/R injury in experimental ischemic models of the heart, liver, brain, and kidney [2-5]. TLRs are a family of highly conserved pattern-recognition receptors (PRR) and are an integral part of the innate immune system. Complementary to the adaptive immune system, the innate immune system is responsible for recognizing conserved molecular patterns via nonclonal PRRs, thus, enabling danger signalling to the cell or the tissue [6]. Bacterial lipopolysaccharides and peptidoglycans as well as fungal zymosan are among the molecular motives recognised by TLRs. However, it important that certain TLRs – namely TLR2, TLR4, and TLR9 – are activated by endogenous ligands, such as ROS, heat shock proteins, and other metabolites, the so-called damage-associated molecular patterns that can accumulate during ischemia [6, 7].

Recent observations suggest that in the kidney, TLR2 and TLR4 are also expressed on tubular epithelial cells (TECs) [8]. Endogenous ligands such as heat shock protein 70 and ROS that can activate TLR2 and TLR4 are strongly up-regulated upon I/R in TECs, and expression of TLR2 and TLR4 is also enhanced [8, 9]. Both TLR2 and TLR4 seem to play an important role in the pathogenesis of I/R injury, as gene-targeted mice lacking either one were protected against renal I/R [10, 11].

Upon ligand recognition, TLRs are activated and initiate a proinflammatory response either through activation and nuclear translocation of nuclear factor kappa-B (NF-κB) [12] or through production of interferon (IFN)-β and IFN-β-inducible genes [13]. NF-κB activation leads to production of several cytokines and chemokines, of which tumor necrosis factor (TNF)-α and interleukin (IL)-8 seem to have key roles in triggering either inflammation or apoptosis. However, it is not known whether deletion of both receptors would have an additive or synergistic protective effect on ischemic tissue injury.

In this study, we determined whether the double genetic deletion of TLR2 and TLR4 would have an additional protective effect in an experimental renal I/R injury model compared with a single gene deletion of TLR2 and TLR4, respectively.

Methods

Mice strains

C3H/HeN and C57BL/6 mice were purchased from Charles River (Sulzfeld, Germany). The generation of TLR2$^{(-/-)}$-, TLR4$^{(-/-)}$-, and TLR2/4$^{(-/-)}$-deficient mice was as follows: TLR2- and TLR4-deficient mice were generated by the 3' end of the extracellular domain, and the major part of the transmembrane region of TLR2 and TLR4 was deleted and replaced by gene targeting with a neomycin-resistant cassette oriented in reverse direction. Homologous recombination was verified by Northern blot and genomic polymerase chain reaction (PCR) analysis. We bred back 129/Sv×C57BL/6 TLR2$^{-/-}$ and 129/Sv×C57BL/6 TLR4$^{-/-}$mice against the genetic C3H/HeN background for five generations to obtain C3H/HeN TLR2$^{-/-}$ and C3H/HeN TLR4$^{-/-}$mice. These were crossed to generate C3H/HeJ TLR4$^{-/-}$/TLR2$^{-/-}$ double-deficient mice lacking TLR2 and TLR4 expression. All mice strains were bred in our own facility. The principles of the National Institutes for Health (NIH) Guide for the Care and Use of Laboratory Animals and the German Law on the Protection of Animals were followed.

Induction of kidney I/R injury

Mice were anesthetized using a combination of isoflurane (Abbott, Germany) inhalation and subcutaneously applied buprenorphine (Essex Pharma, Germany). During the entire experimental protocol, body temperature was maintained at 37°C by placing the animals on a heating pad. After median laparotomy, the aorta and both kidneys with corresponding blood vessels were carefully prepared. Left renal artery and vein were clipped by a microaneurysm vascular clamp for 30 min; meanwhile, a right nephrectomy was performed. After termination of ischemia by removal of the clamp, the remaining kidney was inspected for at least 1 min to verify restoration of blood flow. In each group, six animals were studied.

Harvesting All animals were sacrificed 24 h after induction of left renal ischemia. Blood samples were collected, and kidney tissues were divided to be either snap frozen for cryostaining for protein and RNA analysis or to be fixed in 4% paraformaldehyde for subsequent paraffin embedding.

Functional measurements

Body weight of mice was measured prior to the induction of renal I/R injury. Blood urea nitrogen (BUN) levels were analyzed by an automatic laboratory analyzer (Synchron CX5, Beckman Coulter, Krefeld, Germany).

Morphology

Paraformaldehyde-fixed, paraffin-embedded tissue sections of I/R kidneys were stained with periodic acid-Schiff (PAS) and hematoxylin and eosin (H&E). Analysis was performed by two observers independently in a blinded fashion, and tubular damage was scored per section in at least 20 nonoverlapping fields of view (×400 magnification). Results were scored as following: 0, no tubular damage; 1, area of tubular damage <25%; 2, damage 25–50%; 3, damage >50 %. Loss of brush boarders from proximal tubular cells, tubular cell swelling, interstitial edema, tubular dilation, luminal hyaline casts, and epithelial cell necrosis were used as evidence of tubular damage.

Terminal deoxynucleotidyl transferase-mediated dUTP nick end-labeling (TUNEL)

After incubation of paraformaldehyde-fixed (4 %) sections with Triton 0.1 % and sodium citrate 0.1 %, sections were incubated with TUNEL solution containing terminal deoxynucleotidyl transferase (Boehringer Mannheim, Mannheim, Germany). Following washing and incubation with a rabbit antidigoxigenin antibody (Boehringer Mannheim, Mannheim, Germany), kidney sections were stained using fast red chromatin solution. Positive controls were treated with DNAseI and processed as described above, whereas negative

controls were incubated with phosphate-buffered saline (PBS) instead of TUNEL solution. All sections were counterstained with hematoxylin. Evaluation for positive tubular epithelial cells was performed at magnification of 100× and was related to the number of fields of view per section.

Immunocytochemistry for NF-κB p65 nuclear translocation

Snap-frozen kidney sections of the four different mouse strains were kryosected into slides and fixed with acetone (4°C). After washing with TBS, slides were blocked with 10% protein block (DAKO, Germany) for 1 h and then incubated with rabbit anti-mouse p65 antibody (dilution 1:100) for 30 min at room temperature. Slides were then incubated with a biotinylated anti-rabbit immunoglobulin (Ig)G antibody (dilution 1:100). Positively stained nuclei were evaluated in a blinded fashion by two independent observers and related to the positively stained cytoplasms. Results are shown as percentage.

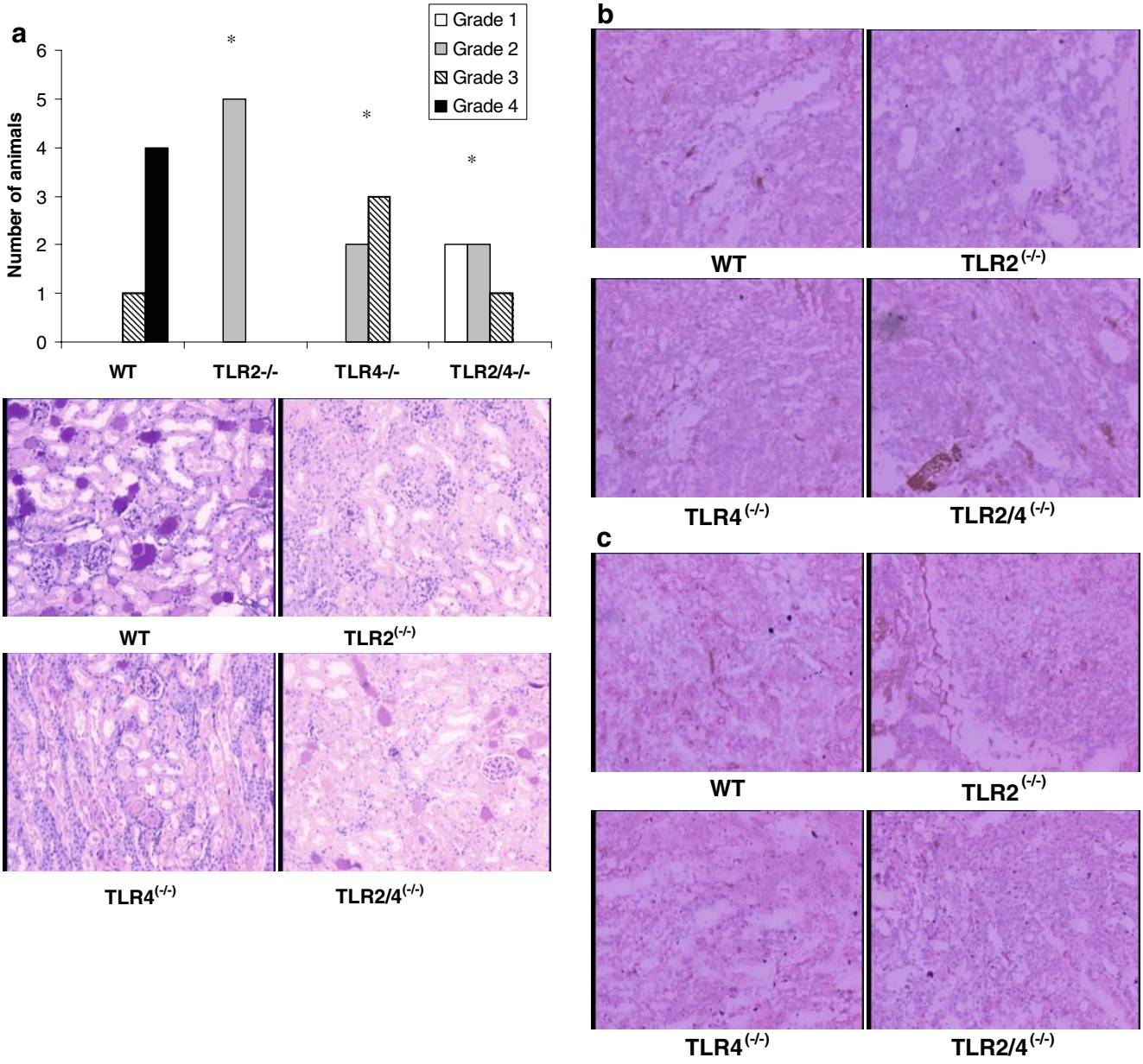

Fig. 1 **a.** Tubular injury in Toll-like receptors (TLR)2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ kidneys was significantly ameliorated than that seen in kidneys from wild-type (WT) mice. Representative sections of kidneys from WT, TLR2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ mice 24 h after reperfusion periodic acid-Schiff (PAS)-stained. *P<0.01 vs. WT. **b** Representative pictures of CD5 staining. **c** Representative pictures of CD68 staining

Immunohistochemistry

Immunohistochemical studies were performed on frozen sections fixed in acetone (4°C). Sections were incubated with mouse primary monoclonal antibodies against the macrophage marker CD68- (clone ED1) and CD5-positive T cells (clone OX19) (Biozol, Eching, Germany), followed by washing the sections and incubation with a secondary rabbit anti-mouse antibody (Dako A/S, Glostrup, Denmark). After washing, sections were incubated with alkaline phosphatase antialkaline phosphatase (APAAP) complex (Dako A/S, Glostrup, Denmark). Binding of antibodies was visualized by development with Fast Red chromogene solution (Dako A/S, Glostrup, Denmark). CD68 and CD5 immunoreactive cells were counted (fields of view/section at 400×).

Cell culture

Primary mouse proximal tubular epithelial cells (PTECs) were cultured from collagenase-digested fragments of proximal tubules obtained from mouse kidneys. Briefly, kidneys were minced and incubated with 0.5 mg/ml collagenase II in Hanks' solution for 1 h. After removing large undigested fragments by sieving (150 µm and 70 µm), the 70-µm sieve yielded a large number of long proximal tubule (PT) fragments without substantial contamination of other nephron segments or glomeruli. The suspension was mixed with an equal volume of 10% fetal bovine serum (FBS) in Hanks' solution and then centrifuged at 1,000 rpm for 5 min at room temperature. The pellets were washed three times with Dulbecco's modified Eagle's medium (DMEM) by centrifugation. Cells were then suspended in growth medium, a mixture of DMEM and Ham's F-12 (1:1) supplemented with 5% FBS, 2 mmol/l L-glutamine, 100 µg/ml epidermal growth factor, 15 mM hydroxyethylpiperazine ethanesulfonic acid (HEPES), 5 µg/ml transferrin, 5 µg/ml insulin, 50 mM hydrocortisone, 50 nmol/l selenium, 0.68 ng/ml 3,3,5 triiodothyronine, 500 U/ml penicillin, and 50 mg/ml streptomycin. These cells have previously been identified as being predominantly of proximal tubular origin. Cells were propagated at 37°C in a humidified-air-containing 5% carbon dioxide (CO_2) incubator. After 7 days, cell cultures were organized as a confluent monolayer.

In vitro hypoxia

PTECs were seeded out and plated on 10-cm plates at 4×10^5/ml 24 h before induction of hypoxia. For hypoxia, the medium was changed to serum-free culture medium saturated with 95% N_2/5% CO_2, and cells were placed in a 37°C airtight box for 24 h. For normoxic conditions, culture medium was changed to serum-free medium, and cells were placed in a 37°C/5% CO_2 incubator for 24 h. After 24 h of hypoxia, cells were analyzed.

IL-8 and TNF-α enzyme-linked immunosorbent assay (ELISA)

Levels of IL-8 and TNF-α in the culture medium were assessed by commercially available enzyme-linked immunosorbent assay (ELISA) kits according to the manufacturers' (R&D Systems, United Kingdom) protocols.

Statistical analysis

Data are expressed as mean ± standard error of mean (SEM). Histological changes were analyzed using the Kruskal–Wallis test, followed by the Mann–Whitney *U* test. Parametric data were compared using one-way analysis of variance, followed by the Bonferroni post hoc test. $P<0.05$ was considered significant.

Results

In vivo experiments

Absence of TLR2, TLR4, and TLR2/4 ameliorates tissue damage and improves kidney function after I/R

The grade of acute tubular necrosis was significantly lower in TLR2/4$^{(-/-)}$ double knock-out mice compared with wild-type mice ($P<0.001$). Both TLR2$^{(-/-)}$ and TLR4 $^{(-/-)}$ mice were protected from ischemic injury, as well, compared with wild-type animals ($P<0.01$). There was a trend between the single knock-out and double knock-out animals showing that ischemic tissue injury might be ameliorated in the double knock-out mice. However, the trend did not reach the level of significance (Fig. 1).

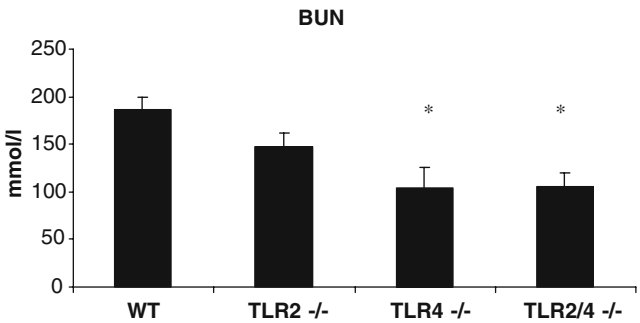

Fig. 2 Blood urea nitrogen (BUN) was significantly lower in Toll-like receptors (TLR)4$^{(-/-)}$ and TLR2/4$^{(-/-)}$ mice versus wild-type (WT) mice 24 h after reperfusion. *$P<0.05$ vs. WT

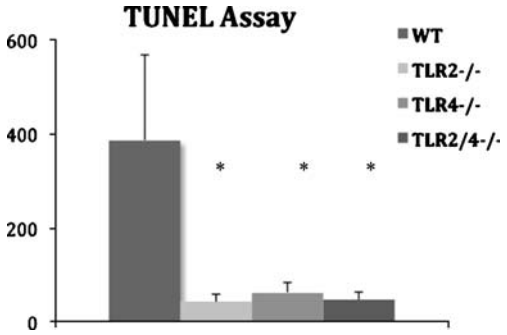

Fig. 3 Number of apoptotic tubular cells 24 h after reperfusion was significantly reduced in Toll-like receptor (TLR)2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ mice compared with wild-type (WT) mice detected by terminal deoxynucleotidyl transferase-mediated dUTP nick-end labeling (TUNEL) staining. *$P<0.001$ vs. WT

BUN levels were significantly lower in TLR2/4$^{(-/-)}$ and in TLR4$^{(-/-)}$ mice compared with wild-type animals ($P<0.05$), and no difference was detected between the TLR2/4$^{(-/-)}$ and TLR4$^{(-/-)}$ groups. Furthermore, no difference was detected in TLR2$^{(-/-)}$ mice versus wild-type controls (Fig. 2).

Apoptosis

TLR2$^{(-/-)}$ and TLR4$^{(-/-)}$ knock-out mice and TLR2/4$^{(-/-)}$ knock-out mice had significantly lower numbers of apoptotic tubular cells in the kidney tissue compared with wild-type litter mates ($P<0.001$). However, there was no significant difference among the knock-out mice (Fig. 3).

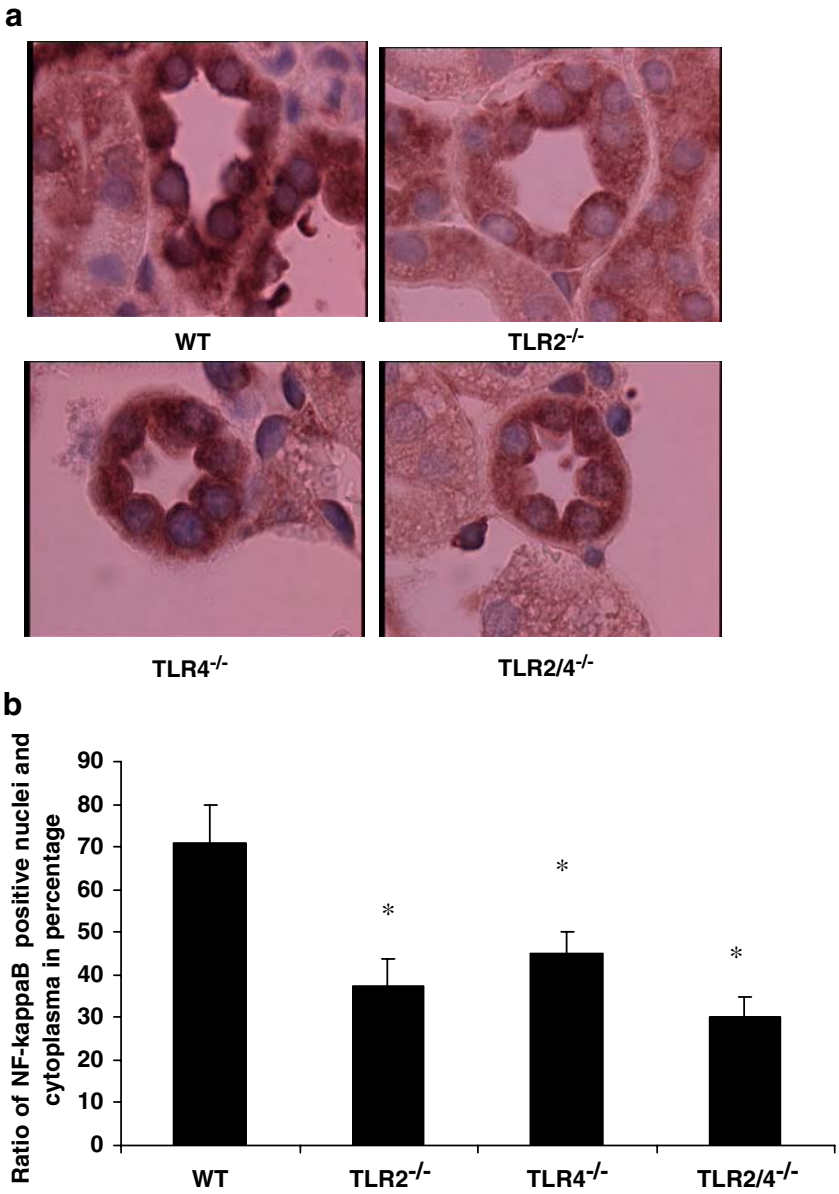

Fig. 4 Translocation of nuclear factor kappa-B (NF-κB) to the nuclei of tubular cells was significantly ameliorated in Toll-like receptor (TLR)2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ mice vs. wild-type (WT) mice 24 h after reperfusion detected by immunochemistry. **a** Representative immunohistochemical staining showing NF-κB translocation into tubular cell nuclei. **b** Percentage of the positively stained cytoplasms. *$P<0.05$ vs. WT

NF-κB nuclear translocation

Nuclear translocation of NF-κB was semiquantitatively analyzed per immunohistochemical staining. Translocation of NF-κB to the nucleus of tubular cells was markedly attenuated in TLR2/4$^{(-/-)}$ mice and was also moderated in TLR2$^{(-/-)}$ and TLR4$^{(-/-)}$ animals versus wild-type mice (Fig. 4).

Inflammatory infiltrates

There was no significant difference in the number of infiltrating CD5- and CD68-positive cells between the experimental groups, as demonstrated by immunohistochemistry (Table 1). Furthermore, there was no difference between the single knock-out mice compared with the wild-type mice.

In vitro experiments

To further investigate the effects of genetic deletions of TLR2 and TLR4 on the inflammatory response under ischemic conditions, we exposed PTECs isolated from knock-out and wild-type mice to 24-h hypoxia. Production of both TNF-alpha and IL-8 was significantly lower in cells isolated from knock-out animals when compared with cells from wild-type mice, detected by ELISA in the cell supernatant (Fig. 5a–b).

There was no difference between cells from TLR2$^{(-/-)}$ and TLR4$^{(-/-)}$ mice, whereas TNF-alpha and IL-8 production were significantly higher in the TLR2/4$^{(-/-)}$ mice compared with the single knock-out animals (Fig. 5a–b).

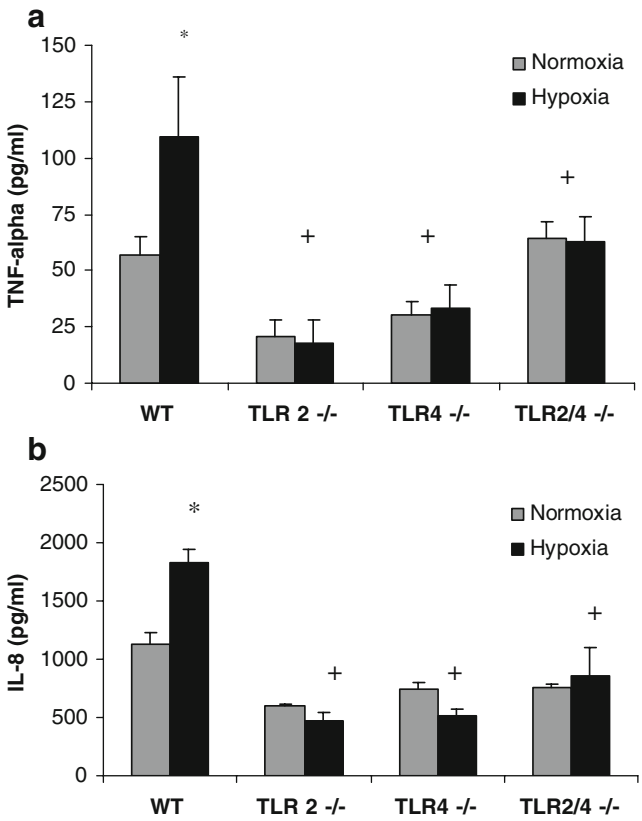

Fig. 5 Tumor necrosis facor (TNF)-alpha and interleukin (IL)-8 production in primary tumor epithelial cells (PTECs) from Toll-like receptors (TLR)2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ and wild-type (WT) mice submitted to 24 h hypoxia in vitro. **a** TNF-alpha and **b** IL-8 production was significantly lower in knock-out mice compared with WT mice. Data are representative of three experiments. *$P<0.01$ vs. normoxic WT, $^{+}P<0.05$ vs. hypoxic WT

Discussion

Our major finding that mice lacking either TLR2 or TLR4 were protected from renal I/R damage as demonstrated by a lower number of apoptotic cells is in accordance with previous findings [11, 12]. Mice lacking either TLR2 or those treated with TLR2 antisense oligonucleotides are protected from renal ischemic injury, both functionally and structurally [11], possibly via decreased production of cytokines and chemokines and thus reduced neutrophil infiltration and apoptosis [11]. Recently, activation of TLR4 has also been demonstrated in a murine ischemic acute kidney injury model using TLR4 knock-out mice [12]. TLR4-deficient mice developed less tubular injury and a more preserved renal function compared with wild-type mice upon I/R injury in the kidney [12]. Furthermore, knock-out mice engrafted with wild-type hematopoietic cells had significantly less azotemia and tubular damage compared with wild-type mice reconstituted with TLR4$^{(-/-)}$ bone marrow, suggesting that signalling through TLR4 in the intrinsic kidney cells plays the dominant role in mediating renal ischemic damage [12].

Table 1 The number of CD5- and CD68-positive cells per field of view in Toll-like receptor (TLR)2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ mice vs. wild-type (WT) mice 24 h after reperfusion, as detected by immunochemistry

	CD5+ leukocytes/field of view	CD68+ leukocytes/field of view
Control	1.1±0.8	0.5±0.12
WT	3.4±2.07	4.2±1.44
TLR2$^{-/-}$	5.2±2.28	5.8±2.28
TLR4$^{-/-}$	5±4.14	5.5±4.18
TLR2/4$^{-/-}$	4.2±1.3	3.8±1.79

We demonstrated that a double genetic deletion of TLR2 and TLR4 does not have an additional protective effect on tissue injury and functional capacity of the kidneys, as determined by serum BUN after renal I/R. The major finding of our study was that TLR2/4$^{(-/-)}$ double-knock-out animals had significantly reduced tissue damage and apoptosis compared with wild-type mice to an extent equal to that in TLR2$^{(-/-)}$ or TLR4$^{(-/-)}$ knock-out mice.

To further investigate whehther there was any difference in the inflammatory response in the kidneys of double-knock-out mice compared with single-knock-out mice, we analyzed the nuclear translocation of NF-κB in vivo and generation of TNF-alpha and IL-8 in vitro. NF-κB translocation was markedly reduced in all TLR-knock-out mice; however, genetic deletion of both TLR2 and TLR4 did not further alter NF-κB activation compared with single-knock-out mice. This result was supported by in vitro data as well, where we demonstrated that generation of TNF-alpha and IL-8 after hypoxia in PTECs of TLR2$^{(-/-)}$, TLR4$^{(-/-)}$, and TLR2/4$^{(-/-)}$ mice was lower when compared with wild-type cells. With respect to in vivo data, double genetic deletion of TLR2 and TLR4 had no further effect on TNF-alpha and IL-8 production. Furthermore, we detected a significant decrease in the rate of apoptosis in all knock-out groups compared with wild-type controls. However, there was no difference between the diverse knock-out groups. These data suggest that TLR2 and TLR4 contribute to the pathogenesis of ischemic organ damage through reinforcement of both inflammation and apoptosis.

TLRs are type-1-membrane-spanning receptors activating similar general signaling pathways but can also trigger its specific pathways. TLR2 and TLR4 can both associate with the adaptor molecule myeloid differentiation primary response gene (88) (MyD88), resulting in activation of mitogen-activated protein (MAP) kinases and phosphorylation of the inhibitor of kappa-B (IκB) kinase, thereby promoting NF-kB translocation to the nucleus and gene transcription. On the other hand, TLR4 but not TLR2 can activate a MyD88-independent toll/interleukin-1 receptor (TIR) domain-containing adaptor protein, inducing an IFN-β (TRIF)-dependent pathway without association of MyD88. Upon activation, TRIF then mediates phosphorylation of the transcription factor interferon-regulating-factor 3 (IRF3), which induces production of IFN and costimulatory molecules. It is surprising that although TLR2 and TLR4 are able to trigger different signaling pathways, we could not detect additional effects when both receptors were genetically deleted. This result indicates that TLR4-mediated MyD88-independent TRIF signaling might not play a central role in ischemic kidney injury, as NF-κB translocation was demonstrated to be altered in all knock-out groups.

This hypothesis is also supported by the results of Wu et al. [11], who demonstrated that TLR4$^{(-/-)}$ knock-out mice were protected from kidney damage following I/R to an extent only equal to MyD88-deficient mice, suggesting that only the MyD88-dependent TLR4 pathway is responsible for TLR4-mediated kidney I/R injury. The impact of TLR4 deficiency on cytokine and chemokine expression, as well as macrophage and neutrophil accumulation, was also consistent with the involvement of the MyD88 pathway. Moreover, the authors found that TRIF-stimulated IFN-inducible gene expression [e.g. interferon-gamma-inducible protein (IP-10) messenger RNA (mRNA) expression] remained unchanged in TLR4$^{(-/-)}$ knock-out mice upon renal I/R injury [12]. Based on our results, we hypothesize that under ischemic conditions in the kidney, TLR2 and TLR4 trigger the same pathways, namely, the MyD88 pathway, leading to NF-κB activation. Therefore, genetic deletion of both receptors has an effect on tissue injury only to an extent equal to TLR2$^{(-/-)}$ and TLR4$^{(-/-)}$-deficient mice.

In conclusion, our study reports the central role of TLR2 and TLR4 in renal I/R injury. However, genetic deletion of both TLR2 and TLR4 does not have an additional effect on tissue injury compared with deletion of TLR2 and TLR4 alone, suggesting that both receptors trigger the same—namely, the MyD88 pathway—under I/R conditions.

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